Ahmed El-Hajjar

Technical Communicator, Product Builder, Software Engineer

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Leadership and Initiatives

Ottawa Computing Group – meetup.com/ottawa-computing-group

***Founder and Host** • 300+ events • 1300+ members • 6+ years every single Friday.* June 2019 – Current

The craziest in-person tech social club in Ottawa, that's not always about tech. No presentations, no agendas, no speakers, no predetermined topics.

- A write-in turned into a lively social. One of the very few tech meetups that remained in-person after the pandemic.
- Attendees share their weekend projects, initiate polarizing debates, and occasionally meet their next employer or collaborator.
- No one else did something as free-form, so I started my own meetup. I'm the eccentric host handling all the trite logistics.

Curiosity-Born Independent Projects

PoetWrite – github.com/cdahmedeh/PoetWrite

Founder and Solo Developer and Writer Open-Source Early Research Project, Documentation Focused, Java

Research-phase word processor for poets. Make rhetorical analysis pain-free so you don't have to count your syllables. Lexicographic assistance to find the right word for your next stanza. Behind-the-scenes technical documentation.

- Born out of frustration that existing poetry apps used slow web APIs and heavy-weight taxing AI. To find out how far poetry analysis and assistance can be done locally before needing any kind of machine-learning, let alone AI.
- Obsession with responsiveness and expandability led to a custom asynchronous MVVM UI framework using reactive patterns, bespoke caching systems and a clever lazy-loading dependency-injected services system.
- To eventually become a fully-fledged product, not a weekend experiment, still solo-developed. Sketches for a UX that make sense, a reassuring QA philosophy with automated tests, and code that I can still understand after taking a 3-week break.

Crosswind – github.com/cdahmedeh/Crosswind

Founder and Solo Developer, Reverse-Engineer Open-Source Compatibility Bridge, .NET C#

Make your forgotten legacy flight simulator work with fancy modern flight planning tools. Clean-room reverse-engineering undocumented protocols that needed way more than just conventional debugging tools.

- I still occasionally use aging decades old flight simulators and wanted to use modern flight planning tools like Navigraph Charts.
- Navigraph's Simlink protocol is not documented but was required for interaction with the source simulator. Conventional debugging tools brought no findings. Detective work using process monitor tools, spying system APIs and inspecting in-memory file structures.
- An executable deliverable, not just stale code on GitHub. Release notes, compatibility lists, build and usage instructions.

TopRoms – github.com/cdahmedeh/TopRoms

***Founder, Steward, Curator** • 200+ GitHub Stars • 4000+ titles • 60+ platforms • Decade-Long Effort* Solo, Fully Released

Personally curated preservation project focused on historically significant and high-quality retro games. Because full-archival collections just can't fit on your HTPC or gaming handheld.

- Created because full-archival collections like No-Intro or Redump couldn't fit on my HTPC and portable gaming devices. The majority of games were shovelware so I only wanted games worth playing on my systems.
- Painstaking and time-consuming research-driven curation process. Pruning through historical magazines, review aggregators, blog posts, independent content and feedback from the community.
- End-to-end distribution, download infrastructure, emulator-ready formats, out-of-the-box packaging. Not just a spreadsheet or script.

Independent Research and Publications

Automatic Transmission Simulation in Games – automatic.cdahmedeh.net

Why does virtually every driving game fail to simulate automatic transmissions correctly? A hunt for the obscure titles that got it right. Taking you on a ride with video demonstrations.

- Was really fascinated by how automatic transmissions had a sophisticated shifting schedule based on throttle input, that seamed to be mind-reading. Selecting gears for maximum fuel-efficiency and maximum performance at the same time.
- However, as a simulation enthusiast, I've played hundreds of titles in my gaming career. And not a single one of them got it right.
- After even more research, finally obscure titles that got it right started to become part of my video game library. Finding out that little discussion was available on the web, it was time for me to make the compilation myself.
- Go over the basics over how automatics behave, mainly for an audience of video game and automative enthusiasts, not an academic exercise.
- Supplemented with demonstration videos, to show off how each game characterized automatic transmissions. Creeping on brake release, early shifting, kickdown, maximum acceleration and up-shifting on cruise.
- Widely referenced by driving simulation enthusiasts and developers as a resource on transmission modeling. ChatGPT and Claude use my post as a reference.

The Sad Demise of Propulsion Controlled Aircraft – pca.cdahmedeh.net

How come a technology that could have saved hundreds of lives be abandoned by the aviation industry? One that would keep aircraft controllable even during a catastrophic flight control failure.

- Having seen documentaries and reading NTSB reports of several flights crippled by loss of hydraulic systems, rendering the aircraft fully uncontrollable and leading to tragic incidents that killed hundreds of people.
- Rather than start with describing the technical details from the onset, an alternative story-telling approach where incidents became the catalyst for the essay.
- Thaught the concept of throttle-only control in a way that even non-aviation laymen could understand.
- Serendipitously came across a NASA research project from 1993 that began on sketches on a few napkins. That allowed pilots to control and safely land aircraft with complete hydraulic and flight control failure using engine thrust alone.
- Showed the effectiveness of throttle-only control based on a flight simulation in addition to a corporate video from NASA showing how the system worked effectively on their test vehicles.
- A melancholic reflection on how FAA considerations influenced the rejection of a technology that may have prevented several major aviation disasters. Despite interest from the aviation community and airline pilots.

Technical Communications Experience and Ownership

Flybits

Technical Writer, Solutions Team March 2024 – June 2026

Joined a startup that had major knowledge fragmentation issues with sparse documentation and no communication infrastructure. Started a ground-up rewriting of documentation for developers, product and even finance. Became the owner of all writing projects, spurring all initiatives, and became THE technical communicator of the entire company, not just a writer. Fostered relationships with stakeholders, not just impromptu SME interviews. Reducing strain on professional services and support.

Exhaustive rewrite of software development kits, web and mobile. More than just a reference guide.

- Complete rewrite of software development kit documentation for mobile and web platforms.
- Not just references or code samples but also included a full-stack tutorial that went through the entire development life-cycle.
- Novice developers could go step by step through the tutorials. Experienced developers could jump straight to the references.
- There was very little documentation for the SDK, and desperately out-dated.
- Required actual coding to develop code samples that actually worked. Had to learn new programming languages, Kotlin and Swift, and become familiar with mobile device platforms, iOS and Android.

- Approach was fostering relationships with engineers for more effective knowledge transfer. Understand the motives behind their design decisions and general usage of the SDK. And my own research by code reverse-engineering.
- Professional services and support felt a significant reduction on strain from professional services and support. Now, they could just point to specific sections in the documentation when working with customers.

Designing and composing a User Guide for Flybits front-end web apps from scratch. Destined to non-tech-savvy readers, like marketers or product owners. Total ownership of the project.

- Virtually no documentation existed for their current web apps except for out-dated documentation for deprecated versions of their tools.
- Developed deep ties with product, support and professional services. Not only for knowledge transfer, but also understand their pain points when supporting their clients to know which areas to focus on.
- Having never used the tools myself, I had to learn the features, inside-out and with depth. So that I could document as a user with production-grade scenarios.
- Primary audiences had limited technical backgrounds, including marketers and product owners. Documentation was catered to them through more exhaustive workflows.
- Documentation structure was inspired by large-scale enterprise software user guides and the progressive learning approach found in flight operating manuals.
- Awareness that many end-users and marketers are reluctant to read documentation, leading to a simplified structure with extensive visual aids, animations, strong searchability using SEO-esque techniques, and step-by-step explanations and navigation.

Onboarding documentation for training accounting. Required after a complete revamp of financial systems. Having to learn a field totally outside of my domain in a short amount of time.

- Flybits underwent a major transformation of its finance systems like handling billing, reimbursements, payroll, and banking.
- The redesigned processes were spurred by the finance leads. They needed to train their accountants on the new systems.
- Had little prior experience with financing and accounting systems. I was required to rapidly learn these unfamiliar concepts and terminology. Deadlines were incredibly tight so that accounting activities wouldn't be interrupted.
- It was even more difficult because I didn't have full access to their accounting or banking software for confidentiality reasons. Knowledge transfer sessions were very long and during the accounting processes as they happened, not just simulated ones.
- Documentation was structured into a hybrid style with a heavy focus on UX. Was presented with workflows for each accounting activity, but also the ability to jump into a specific section to reference a particular task.

Syntronic

Technical Writer, Business Analyst May 2022 – November 2023

Developing a plan to implement software that fit customers' needs for software and services. Done through an analysis process taking in formal requirements into software specifications. Domains include internal research projects, machine-learning initiatives, sterilization medical labs, embedded systems and software development. Company-wide reviews of developer documentation, meaning a constant learning process across embedded, automotive, telecom, medical, IoT, machine learning and AI domains.

Requirements analysis contributor for an automated medical sterilization lab.

- Worked with other writers and product developers for requirements-gathering sessions to find out the technical needs of a newly developed automated medical sterilization lab.
- Search for ambiguities, contradictions and missing information in requirements documentation. To further refine docs by working with other writers and the clients themselves.
- Transformed requirements into software specifications and implementation plans for engineering teams. Along with roadmaps for future development projects.
- Wrote user stories, acceptance criteria, and test cases to support development and help quality-assurance team with their testing process.

Automating requirements management with continuous integration tools and documentation platforms.

- Requirements, specifications, and implementation plans were written mainly in Microsoft Word and PDF documents, resulting in significant manual efforts to transfer them into tickets for Azure DevOps.
- Created an application, that wasn't just a script, to automatically extract requirements and acceptance criteria from the aforementioned requirements documents. It would automatically create equivalent tickets in their Azure DevOps repo.
- Eliminated the need for manual data entry, that was primarily done by the QA team. Also ensured our requirements to be equivalent with the test cases that QA were going to verify. Meaning their testing matched our requirements.

Company-wide reviews and maintenance of engineering documentation for several engineering teams. Required learning of several unfamiliar fields.

- Syntonic is an R&D heavy organization. With engineers in a variety of fields, generally embedded systems, for automotive, telecom, medical, defence, IoT and recently, machine learning and AI.
- Engineering teams often wrote the technical documents themselves, but I constantly found major shortcomings. Mainly a lack of explanations of topics that the engineering teams took for granted but was needed for someone outside the field to understand.
- Nearly every review required me to learn a new domain, new frameworks, and especially new technologies very quickly. This was mainly to understand the intent and content of the documentation, but also fill in the knowledge gaps I mentioned above.
- Improving UX of documentation, restructuring formatting, consistent presentation and easier navigability. Along with formal readability analysis to adjust wording, sentence structure and frequent rewriting to make documentation easier to read for their target audience.
- Even validated code and fixed typos in command line instructions. And checked that links didn't point to deprecated or broken pages.

Amazon, Entrust, 1VALET

Technical Writer, Consultant October 2020 – October 2022

Focused on complex systems that were poorly documented. For robotics platforms, automation infrastructure, web security tools, property technology. Looking into source code, interviews with SMEs, and tracing actual workflows rather than just theoretical test cases. With that, produce onboarding guides, user docs, success guides, knowledge-transfer documentation. Targeted both for engineers and for support who don't have much technical fluency.

Software Engineering Career (More Than 10 Years)

Cisco, ESDC Service Canada, Ford Motor Company

Software Consultant February 2019 – October 2020

Consultation for modernization of legacy enterprise systems to modern platforms. Software engineering, automation, deployment, testing, and architecture initiatives. In telecommunications, government, and automotive sectors. Documentation for cloud infrastructure and deployment, continuous integration and deployment, automated testing, technical documentation.

Lixar, NorthStar, ShopLiftr, MediaVantage, Teldio

Software Developer May 2011 – February 2019

Full-stack software developer authoring backend cloud platforms, APIs, mobile applications, enterprise applications, machine learning tools, software for IoT devices. And research and concepts for emerging technologies. Responsible for architecture, backend development, DevOps, documentation and product support. Servant leadership and mentoring junior developers.

Skills

Domains and Industries

Financial Technologies, Healthcare, Medical Technology, Laboratory Automation, Robotics, Cybersecurity, Telecommunications, Networking, VoIP, Property Technology, Government, Public Sector, Automotive, Retail Technology, Marketing Technology, Utilities, Cloud Services, Enterprise Software, Mobile Applications, Web Applications, Modeling and Simulation

Documentation and Knowledge Management

Developer Documentation, SDK Documentation, API Documentation, Technical Documentation, User Guides, Success Guides, Deployment Documentation, Support Documentation, Knowledge Transfer, Release Notes, Changelogs, Version History, Specifications, Requirements, User Stories, Test Cases, Prototyping, Information Architecture, Technical Communication

Emerging Technology and Research

Artificial Intelligence, Machine Learning, OCR, Blockchain, Bluetooth, Proximity Technologies, Software-Defined Radio

Business Analysis and Product Discovery

Requirements Gathering, Requirements Engineering, Specifications Development, Customer Interviews, SME Interviews, Business Analysis, Use Cases, Workflow Analysis, Process Mapping, Acceptance Criteria, Product Ownership, Product Support, Code Exploration, Visualization

Leadership and Collaboration

Project Management, Stakeholder Management, Interdepartmental Communication, Mentoring, Peer Reviews, Servant Leadership, Event Hosting, Moderation, Knowledge Transfer

Visual Modeling and Design

UML, Sequence Diagrams, State Diagrams, Flowcharts, Swim Lanes, Mind Maps, Wireframing, Balsamiq, Microsoft Visio, draw.io, User Experience (UX)

Content Creation and Publishing

Whitepapers, Case Studies, Blogging, Ghostwriting, Copywriting, Editing, Research, Fact Checking, Marketing Communications, Announcements, Social Media, Content Strategy

Documentation and Authoring Tools

GitBook, Confluence, MediaWiki, SharePoint, Google Docs, Microsoft Word, Scriviner, Markdown, Antidote

Artificial Intelligence (AI)

Prompt Engineering, LLM Training, Hugging Face, LM Studio, OpenAI ChatGPT, Anthropic Claude, Google Gemeni

Spoken Languages

English, French, Arabic, Cat (Domestic Fluency)

Programming Languages

Java, C#, Python, JavaScript, Swift, Kotlin, C++, Go, PHP, Ruby, Visual Basic

Scripting Languages

bash, PowerShell, AppleScript

Version Control and Source Control

Git, GitHub, GitLab, Bitbucket, SVN, CVS, Branching, Merging, Tagging, Pull Requests, Code Reviews, Release Management

Web Development

HTML, CSS, JavaScript, Angular, React, Vue.js, Single Page Applications, Server-Side Rendering

Desktop Applications

Swing, SWT, Eclipse SDK, WinForms, JIDE

Game Development

Clickteam Fusion (formerly Multimedia Fusion), Construct

Software Development and Architecture

Object-Oriented Design, Functional Programming, MVC, MVVM, Dependency Injection, Multi-Threading, Asynchronous Programming, Reactive Programming, Reverse Engineering, Test-Driven Development, Software Architecture

Mobile Development

Android, iOS, Firebase, APNs, Retrofit, Mobile SDK Integration

Backend Development

Java EE, Spring, Spring Boot, Django, Ruby on Rails, REST APIs, SOAP, Swagger, Postman

Database and Data

SQL, PostgreSQL, MySQL, SQL Server, Oracle, SQLite, Hibernate, JPA, Liquibase, Flyway

DevOps and Platforms

Azure DevOps, GitHub, GitLab, Bitbucket, Jenkins, TeamCity, Bamboo, Docker, Docker Compose, CI/CD, Maven, Gradle

Operating Systems and System Administration

Windows, Linux, macOS, SSH, Windows Server, Package Management, System Administration, Virtualization, Backup Systems

Cloud and Infrastructure

Amazon AWS, Microsoft Azure, DigitalOcean, S3, CloudWatch, Redshift, DynamoDB

Embedded and Hardware

Embedded Systems, FPGA, Arduino, Raspberry Pi, Circuit Design, PCB Components, Soldering, 3D Printing

Additional Experience

Uber

Partner and Driver *• Part-Time • 4.9 Star Rating • 1000+ trips* January 2019 – September 2025

Ride-sharing side-gig as a passenger driver during late nights and weekends. Was to immerse myself into a customer service business and expose myself to blue-collar work. Helped me develop communication and interpersonal skills that could be applied in other areas in my field.

Volunteering

Ottawa Stray Cat Rescue

Lead Volunteer *• Occasional* February 2020 – Current

Organizing fundraisers and adoption events for the local animal rescue for stray and feral cats throughout Ottawa.

Selected Essays

Electric Youth -- Marnie – marnie.cdahmedeh.net

Analytical essay exploring music theory, lyrics, themes, visual storytelling, and artist's intent for a contemporary synth-pop song.

Education

Studies in Software Engineering, Computer Engineering, Computer Science

University of Ottawa September 2009 – December 2014

Studies in Technical Communications

Simon Fraser University (Online) June 2018 – November 2018